

Markus Andreas Otsmaa

+372 5014700 | Eindhoven, NL
markus.otsmaa@gmail.com | [linkedin.com/in/markus-otsmaa](https://www.linkedin.com/in/markus-otsmaa) | markusotsmaa.com

PROFILE

As a mechanical engineering student at TU/e, I combine technical curiosity with the mindset of a professional athlete. I'm highly driven, goal-oriented, and committed to strong teamwork. With a high average grade of 8.3/10, I pair academic performance with a very high stress tolerance. I am interested in how emerging technologies in manufacturing can be applied to create safer, more reliable, and more efficient systems.

PROJECTS

- Propeller Optimiser** | *Personal Project; Python, OpenFOAM, CalculiX* June 2026 to Sept. 2026
- Multi-Fidelity Pipeline:** Python pipeline generating Pareto-optimal propeller geometry from motor specs, chaining BEMT, OpenFOAM CFD and CalculiX FEA under NSGA-II optimisation.
 - CFD Validation:** Validated the chain against APC reference data; split the residual thrust deficit into a hover-only term and a term traced to the NACA 4412 proxy for the unpublished APC section.
 - Root-Cause Analysis:** Found a blade-placement orientation bug, showed steady MRF cannot sustain a hover slipstream, and ruled out transition modelling.
 - Verification Tooling:** Built a preflight audit flagging unexercised config and silently ignored OpenFOAM options; characterised the noise floor (4× iterations shifted C_T by 3.6%).
- Generative Design in the Automotive Industry** | *Thesis; Fusion 360, FDM* Sept. 2022 to May 2023
- Generative Design Setup:** Redesigned LRP Twister S10 RC buggy rims in Fusion 360 generative design: preserve/obstacle geometry, one 60° sector patterned into a 6-spoke rim, pin-constrained hub, 6 N rim load, safety factor 1.2, mass minimisation.
 - Design for Additive Manufacturing:** Ran the solver against additive, die-cast and unconstrained method sets, then FDM printed the selected AM candidates, producing organic spoke geometry not achievable by conventional machining.
 - Failure-Driven Iteration:** First printed rims fractured in a crash; raised the design load to 9 N and reprinted in a more ductile PLA, giving rims 52 to 57% lighter than printed copies of the stock rim (a 25.7% reduction in rear wheel mass incl. tyre).
 - Instrumented Testing:** Photogate timing rig (LabQuest 2, 1 ms); over 12 runs per rim set the generative rims cut mean 1.9 m run time 5.8% vs. printed stock-rim copies (0.611 s vs. 0.649 s).
- Shuffleboard Robot** | *Group Project, Mechatronic Design (4CC10)* 2026
- Drivetrain & Sensing:** Verified motor and gearbox sizing and selected sensors (VL53L1X, Baumer OM30-P).
 - Design Documentation:** Produced FMEA risk tables and functional-architecture diagrams in $\text{\LaTeX}$.

EDUCATION

- Eindhoven University of Technology** Sept. 2024 to Present
Bachelor of Science in Mechanical Engineering Eindhoven, NL
- Average Grade:** 8.3/10
 - Relevant Coursework:** Mechatronic Design, Dynamics and Control of Robotics Systems, Dynamics and Control of Mechanical Systems, Control of a Flexible Robot System, Road Vehicle Dynamics, Solid Mechanics, Structure and Properties of Materials, Multiped Robot, Design of a Launching Mechanism, Computational Mechanics (current).
- Tallinn Secondary School of Science** Sept. 2021 to June 2024
High School, Science and Engineering Track Tallinn, Estonia

SKILLS

- CAD:** SolidWorks (CSWP in progress), Siemens NX, Fusion 360
Simulation: OpenFOAM (CFD), CalculiX, Marc Mentat, Fusion 360 (FEA), BEMT
Manufacturing: FDM additive manufacturing, design for additive manufacturing
Optimisation: NSGA-II multi-objective optimisation, Fusion 360 Generative Design
Programming: Python, MATLAB, Simulink, $\text{\LaTeX}$
Certifications: Certified SolidWorks Associate (Mar. 2026), Cambridge Advanced English C2 (Apr. 2024)
Languages: Estonian (native), English (C2)

ACTIVITIES

- Competitive Ballroom Dancing** | *Elite level, ~15 years* Until 2026
- Estonian Under-21 Champion; top-15 at the WDSF Under-21 World Championship.

REFERENCES

- Andreas Vasarik**, IGCSE Teacher, Collège du Léman, andreas.vasarik@cdl.ch, +41 774837548